

Information loss of fixed linear health indicators under competing degradation mechanisms and variable operating conditions

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Abstract

Predictive maintenance often estimates remaining useful life from a health indicator: a fixed-weight combination of condition-monitoring signals. When a component degrades through several mechanisms at different rates, the combination most informative about remaining life, the informative direction, changes with age, and no fixed indicator can follow it. This paper quantifies the information such an indicator loses and whether the operating point, which sets those rates, can reduce it. For units differing only in age, the loss depends on one angle between the indicator and the informative direction and fixes the increase in the error variance of the remaining-life estimate. If that direction turns through an arc Φ over life, a well-placed fixed indicator increases the variance by at most a factor $1/\cos^2(\Phi/2)$. On 129 lithium-ion cells the arc is 30.7° , so by at most 7.5%, and remaining-life estimates and replacement decisions are as good with a fixed indicator as with all signals. In simulation, adjusting the operating point reduces this already small loss 47-fold but barely improves the estimate, while lifetime varies 19-fold. Unless the arc is large, a fixed indicator is adequate and the operating point is better chosen for lifetime.

Keywords: health indicator, remaining useful life, prognostics, predictive maintenance, competing degradation mechanisms, operating conditions,

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1. Introduction

Predictive maintenance replaces or repairs a component shortly before it is expected to fail, rather than on a fixed schedule or after failure. The expectation comes from an estimate of the component's remaining useful life, and that estimate is in turn computed from condition monitoring. Monitoring seldom passes its raw measurements to the estimation algorithm: channels such as vibration, temperature, impedance and capacity are first combined into a single number, the health indicator, and the algorithm sees only that number [1, 2]. In practice the weights of the combination are usually fixed when the monitoring system is designed, because they are embedded in firmware, frozen by certification, or impractical to re-tune in the field. The indicator therefore determines how accurately the remaining useful life can be estimated and, through that estimate, how early maintenance must be scheduled to keep the risk of failure acceptable.

A fixed weighting is not always a good weighting. Engineering components commonly degrade through several mechanisms at once, and these mechanisms need not progress at the same rate. In a lithium-ion cell, for example, capacity fade and the growth of internal resistance arise from different physical processes and need not evolve in proportion; one mechanism may dominate early in life while another accelerates towards failure. It helps to picture the monitoring channels as the axes of a space. As the component ages, its readings move through this space, and the direction in which they move is the combination of channels that changes most with age, and hence the one that says most about the remaining life. When the mechanisms progress at different rates, this direction turns as the component ages. A fixed indicator is a fixed direction in the same space, so it can be aligned with the most informative combination at one moment only, and the information it discards at other moments cannot be recovered by any prognostic algorithm that uses the indicator [3].

Two practical questions follow. The first is how much information a fixed indicator actually loses, and whether this can be assessed before the indicator is designed, when a different weighting or an adaptive indicator can still be chosen. The second concerns the operating conditions, such as temperature, load or duty cycle. These vary, either because the duty imposes them or because the operator selects them, and in both cases they change the relative

36 rates of the mechanisms and hence the direction in which the readings move.
37 Operators already select operating points to extend life [4, 5, 6]; the question
38 is whether the operating point could also be chosen to keep a fixed indicator
39 aligned, and whether doing so would be worthwhile.

40 Existing work addresses the first question from the perspective of indica-
41 tor construction, with criteria that differ from the one used here. Health indi-
42 cators are constructed from data so that they change steadily and predictably
43 with age, as measured by criteria such as monotonicity and trendability [2, 7],
44 or by fusing sensors into a composite index whose weights are fitted once,
45 for example so that the fused index has a consistent value at failure or fol-
46 lows a prescribed degradation model [8, 9, 10, 11, 12]; such indices have
47 also been built under several operating conditions [13] and learned by neural
48 networks [14], and reviews survey the resulting pipelines [1, 15]. Once fitted,
49 these indices are fixed and linear in the channels, which is precisely the class
50 of indicators considered in this paper, but none of the construction criteria
51 measures how much information about remaining life the index discards as
52 the mechanisms diverge. State-space fusion [16] avoids a fixed weighting but
53 tracks a single degradation state, in which case the direction cannot turn.
54 Section 4 compares one fusion criterion with the design proposed here on
55 measured cells.

56 The second question connects to degradation modelling and to operation.
57 Degradation under time-varying operating conditions is well modelled [17,
58 18, 19, 20, 21, 22, 23], and several degradation characteristics are known
59 to respond differently to the same environment [24] or to depend on one
60 another [25]. A separate body of work selects the operating point, or the
61 load or production rate it implies, to extend life or to balance performance
62 against reliability [4, 5, 6, 26, 27, 28], and prognostic accuracy has been
63 assessed through its consequences for predictive maintenance [29]. These
64 studies treat the monitoring signal as given; none asks what the operating
65 point does to the information a fixed indicator retains. Our previous work
66 bounded the information that any fixed compression of the channels retains
67 along a single, given degradation trajectory [3], and the number of run-to-
68 failure units required to learn such a compression [30]. The present paper
69 builds on that analysis: it expresses the loss of a single linear indicator
70 through one angle, relates the loss to the error of the remaining-life estimate
71 and to maintenance decisions, bounds it by how far the informative direction
72 turns, tests the bound on measured data, and lets the trajectory change with
73 the operating point.

74 The main contributions of this paper are as follows.

- 75 • The information discarded by a fixed linear indicator is shown to be
76 an exact function of a single misalignment angle, and the resulting
77 inflation of the error variance of the remaining-life estimate, and of the
78 safety margin of a predictive replacement policy, is obtained in closed
79 form (Section 2).
- 80 • The arc traversed by the informative direction over the service life is
81 shown to bound this inflation before any indicator is designed, which
82 yields a five-step design procedure that requires run-to-failure data only
83 (Section 3).
- 84 • In a case study on measured battery cells the bound is 7.5%, which
85 corresponds to a safety margin of a predictive replacement policy at
86 most 3.7% wider; a leave-one-out remaining-life estimator and a re-
87 placement policy using the fixed indicator perform as well as ones using
88 all channels (Section 4).
- 89 • Selecting the operating point to keep the indicator aligned reduces
90 the information loss substantially but leaves the error of a practical
91 remaining-life estimator essentially unchanged, whereas the operating
92 point alters the lifetime by an order of magnitude (Section 5). Public
93 run-to-failure data support the premise that the informative direction
94 turns but do not permit a test of the effect of the operating point on
95 degradation (Section 6).

96 The remainder of the paper proceeds from the general to the specific.
97 Section 2 states the degradation model, defines the information loss and lists
98 the assumptions. Section 3 bounds the loss by how far the informative di-
99 rection turns and derives the design procedure that follows from the bound.
100 Section 4 applies the procedure to measured lithium-ion cells, up to a mainte-
101 nance decision. Section 5 examines, on a synthetic system, whether selecting
102 the operating point is worthwhile, and Section 6 examines the assumptions on
103 public run-to-failure data. Section 7 answers the two questions and discusses
104 the implications. Derivations are given in Appendix A.

105 2. Problem formulation

106 This section states how the mechanisms degrade and how they are ob-
107 served (Sections 2.1 and 2.2), defines the information that a fixed indicator

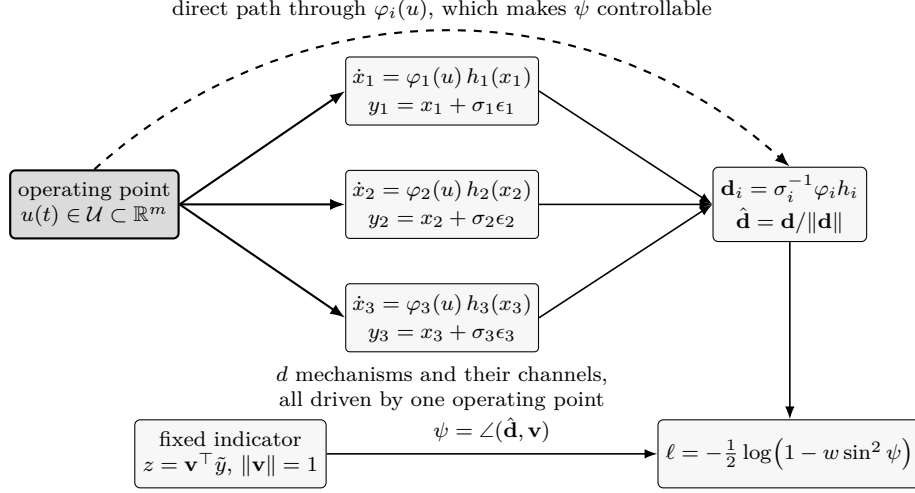


Figure 1: Overview of the problem formulation. One operating point drives all d mechanisms, each observed through its own channel. The informative direction $\hat{\mathbf{d}}$ is the weighting of the whitened channels that responds most strongly to the remaining life; the loss ℓ of a fixed indicator $\mathbf{v}$ depends only on the angle ψ between them. The operating point affects $\hat{\mathbf{d}}$ both through the damage and directly through the stress factors $\varphi_i(u)$; the direct path makes ψ controllable.

108 loses and relates it to the error of the remaining-life estimate (Section 2.3),
 109 and closes with the assumptions on which the rest of the paper rests. Figure 1
 110 gives an overview of the construction.

111 2.1. Degradation model with operating-point dependence

112 A component degrades through d mechanisms, with damage states $x =$
 113 $(x_1, \dots, x_d)$, $x_i \geq 0$. A single operating point $u(t) \in \mathcal{U} \subset \mathbb{R}^m$, comprising
 114 m adjustable variables such as temperature, load and duty cycle, drives all
 115 of them; the envelope $\mathcal{U}$ is the bounded range of operating points allowed in
 116 service. The damage of each mechanism grows at a rate that depends on the
 117 operating point and on the damage already accumulated:

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d. \quad (1)$$

118 The stress factor $\varphi_i(u) > 0$ expresses how much faster mechanism i runs at
 119 u than at a reference operating point, and the shape function $h_i(x_i) > 0$
 120 describes how its rate changes as damage accumulates. The two effects enter
 121 as separate factors, which makes the model separable. This is the standard

122 accelerated-life form [17]: in the Arrhenius model $\varphi_i(u) = A_i \exp(-E_i/(kT))$,
 123 with activation energy E_i , absolute temperature T and Boltzmann constant
 124 k , and suitable choices of h_i yield exponential, power-law and stretched-
 125 exponential degradation [3]. Damage is normalised such that each mechanism
 126 exhausts the component at unity, and the component is withdrawn from
 127 service when the first mechanism reaches a condemnation limit $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (2)$$

128 The lifetime T_f depends on the operating point, and the normalised life
 129 $\tau = t/T_f$ runs from 0 at the start of service to 1 at withdrawal.

130 2.2. Remaining useful life and the informative direction

131 The quantity to be estimated is the remaining useful life $R(t) = T_f - t$.
 132 For a single unit with known parameters, (1) would determine it exactly;
 133 the uncertainty arises because the units of a fleet differ [17]. The analysis
 134 assumes that they differ only in age: all units follow a common degradation
 135 path, and each is at a different position along it, as are identical components
 136 installed at different times. Such a population is called a time-shift popula-
 137 tion (Fig. 2(a)), and R is taken to be Gaussian over it with variance $\text{Var}[R]$.
 138 For two units of such a population whose remaining lives differ slightly, the
 139 damage states differ, to first order, only along the direction in which the
 140 damage is currently growing. Since R decreases at unit rate, the sensitivity
 141 of mechanism i to the remaining life is therefore

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (3)$$

142 Each mechanism is observed through a dedicated channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (4)$$

143 with independent standard normal noise ϵ_i and noise level σ_i . This pairing of
 144 channels with mechanisms is an assumption: condition monitoring in general
 145 yields $y = H(x, u) + \epsilon$, with sensors that mix mechanisms, and Section 6 shows
 146 that raw turbofan sensors violate it. To make channels with different units
 147 and noise levels comparable, each is divided by its noise level, which is called
 148 whitening; the whitened readings $\tilde{y} = (y_i/\sigma_i)$ at one inspection are called the
 149 record. By (3), the record responds to the remaining life through the signal
 150 vector

$$\mathbf{d} = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d} / \|\mathbf{d}\|. \quad (5)$$

151 The norm $\|\mathbf{d}\|$ measures how strongly the record responds to the remaining
 152 life, and the unit vector $\hat{\mathbf{d}}$, a point on the unit sphere of the d -dimensional
 153 space of whitened channels, is the combination of channels along which it
 154 responds. It is referred to as the *informative direction*: the weighting of the
 155 channels that, at the current state, carries the most information about the
 156 remaining life. All entries of $\mathbf{d}$ are positive, and $\hat{\mathbf{d}}$ depends on both the state
 157 and the operating point. The operating point affects $\hat{\mathbf{d}}$ through two paths:
 158 indirectly, through the damage it produces, and directly, through $\varphi_i(u)$.

159 2.3. Information loss of a fixed linear indicator

160 A fixed linear indicator combines the record with a weight vector selected
 161 once at the design stage:

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1. \quad (6)$$

162 The weight vector is a second point on the same unit sphere. An indicator
 163 defined on raw measurements, with weights $\mathbf{a}$, is the same mapping with
 164 $\mathbf{v} \propto \Sigma^{1/2} \mathbf{a}$, where $\Sigma = \text{diag}(\sigma_i^2)$. Let ψ denote the angle between $\mathbf{v}$ and
 165 $\hat{\mathbf{d}}$, called the misalignment, and let $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ denote the signal-to-
 166 noise ratio of the record, that is, the spread of the remaining life across the
 167 population as seen through the record, relative to the noise. Linearising
 168 the record about the nominal path renders it jointly Gaussian with R ; the
 169 indicator then retains the fraction $\cos^2 \psi$ of the signal and all of the noise
 170 (Fig. 2(b)).

171 Information is measured here by the mutual information between the
 172 remaining life and the observation, in natural units (nats), and the loss ℓ is
 173 the information about R carried by the record but not by the indicator. It
 174 is therefore (Appendix A.1)

$$\ell = \frac{1}{2} \log \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (7)$$

175 with $w = \gamma/(1 + \gamma)$. The loss has a direct statistical meaning: the error
 176 variance of the optimal estimate of R based on the indicator exceeds that
 177 based on the record by the factor

$$\frac{\text{Var}[R | z]}{\text{Var}[R | \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}. \quad (8)$$

178 A loss of 0.01 nat, for instance, means an error variance about 2% larger
 179 than with all channels, and this excess, $e^{2\ell} - 1$, is the quantity that the rest

180 of the paper reports alongside the loss. The loss depends on the indicator
 181 only through the single angle ψ , and for small misalignment $\ell \approx \frac{w}{2}\psi^2$. The
 182 design objective is its average over the service life,

$$\bar{\ell} = \frac{1}{T_f} \int_0^{T_f} \ell(t) dt. \quad (9)$$

183 The same reduction applies when the indicator is read at many inspections,
 184 with γ summed over the records of the inspections and $\sin^2 \psi$ averaged with
 185 weights $\|\mathbf{d}\|^2$ (Supplementary Section S2).

186 The loss also bears directly on maintenance decisions. Under the Gaussian
 187 model, a predictive replacement policy that replaces a unit once its proba-
 188 bility of failing within a lead time t_{lead} reaches α does so when the estimated
 189 remaining life falls to $t_{\text{lead}} + \zeta_\alpha s_R$, where s_R is the standard deviation of the
 190 remaining life given the observation and ζ_α the standard normal quantile of
 191 order $1 - \alpha$. The safety margin $\zeta_\alpha s_R$ is, on average, remaining life discarded
 192 at replacement beyond the lead time, and a fixed indicator multiplies it by e^ℓ
 193 relative to the record (Fig. 2(c)). Decision-oriented assessments of prognostic
 194 algorithms weigh precisely this margin against the risk of failure [29].

195 A further consequence concerns the design of tests and sensors. Adding or
 196 improving sensors increases the information in the record, but does not by it-
 197 self reduce the loss of a fixed indicator. The design of accelerated degradation
 198 tests and of inputs for estimability usually maximises the Fisher information
 199 of the record [31], here $I_{\text{rec}} = \|\mathbf{d}\|^2$ about R when the prior variance is
 200 normalised to one; the loss, however, is determined by this quantity together
 201 with the Fisher information of the indicator itself, $I_z = (\mathbf{v}^\top \mathbf{d})^2$:

$$\ell = \frac{1}{2} \log \left(1 + \text{Var}[R] I_{\text{rec}} \right) - \frac{1}{2} \log \left(1 + \text{Var}[R] I_z \right). \quad (10)$$

202 For a fixed value of I_{rec} the loss still ranges from zero to $\frac{1}{2} \log(1+\gamma)$ as ψ varies.
 203 Maximising the information in the record therefore does not minimise the loss
 204 of a fixed indicator, whereas including I_z in the criterion does. Supplementary
 205 Section S4 shows this on the synthetic system of Section 5.2.

206 2.4. Assumptions

207 The analysis rests on four assumptions, to which the later sections refer.

208 (A1) Time shift: the units of a population follow a common degradation
 209 path and differ only in their position along it.

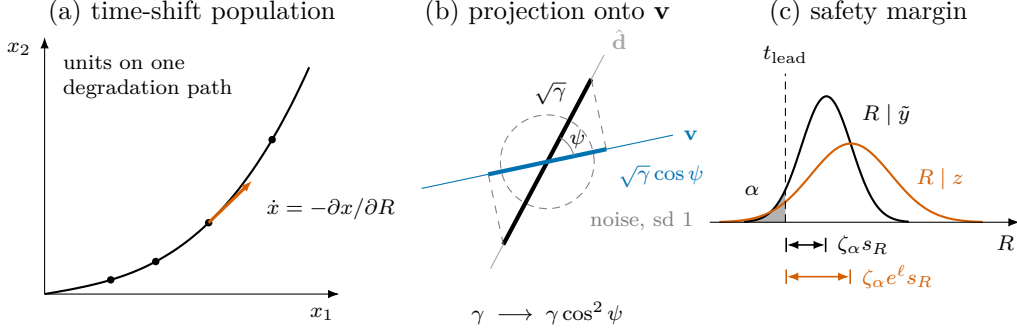


Figure 2: Mechanism of the information loss. (a) In a time-shift population the units follow one degradation path and differ only in their position along it, so neighbouring units differ along the tangent $\dot{x}$, which by (3) is the sensitivity to the remaining life. (b) In the whitened record, drawn for $d = 2$, the spread of the remaining life moves the record along $\hat{\mathbf{d}}$ with standard deviation $\sqrt{\gamma}$, and the noise has unit standard deviation in every direction. Projecting onto $\mathbf{v}$ keeps $\sqrt{\gamma} \cos \psi$ of the signal and all of the noise, so the signal-to-noise ratio falls from γ to $\gamma \cos^2 \psi$, which gives (7). (c) Conditional densities of R when a predictive replacement policy replaces: each has probability α below t_{lead} , so its centre lies ζ_α standard deviations above it. The indicator widens the density by e^ℓ (8) and the safety margin with it; e^ℓ is drawn as 1.6 for legibility.

- 210 (A2) Linear-Gaussian record: the remaining life is Gaussian over the popu-
- 211 lation, and the record is linearised about the nominal path.
- 212 (A3) Channel pairing: each mechanism is observed through its own channel
- 213 with independent Gaussian noise, as in (4).
- 214 (A4) Separable stress: the operating point and the accumulated damage
- 215 affect each mechanism's rate as separate factors, as in (1).

216 The loss (7) and the bound of Section 3 need (A1) and (A2), and hold
 217 for the direction in which the observed channels respond, whether or not
 218 each channel tracks a single mechanism. (A3) and (A4) are needed only
 219 for the analysis of the operating point in Section 5. Section 4 states which
 220 assumptions the measured cells satisfy, and Section 6 tests them on public
 221 data.

222 3. Rotation of the informative direction and a bound on the loss

223 Section 2 showed that the loss of a fixed indicator depends only on its
 224 angle to the informative direction. How large that angle must become over a
 225 life depends on how far the informative direction turns. This section explains

226 why it turns, bounds the loss by the length of its path, and derives from the
 227 bound a design procedure.

228 3.1. Rotation rate of the informative direction

229 As the component ages, the tip of the unit vector $\hat{\mathbf{d}}$ traces a path on the
 230 unit sphere. Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i and
 231 $p_i = \hat{\mathbf{d}}_i^2$ the share of the record's Fisher information carried by channel i .
 232 Differentiating (5) along (1) gives (Appendix A.2)

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g}_i = \varphi_i(u)h'_i(x_i) + \nabla_u \log \varphi_i(u) \cdot \dot{u}, \quad (11)$$

233 where Std_p is the standard deviation over channels weighted by the shares
 234 p_i . The informative direction thus turns at a speed equal to the information-
 235 weighted spread of the logarithmic rates of the mechanisms. If all mechanisms
 236 accelerate in proportion, the direction stays fixed however rapidly the com-
 237 ponent degrades; it turns only because the mechanisms disagree (Fig. 3(a)).
 238 The second term of $\dot{g}_i$ represents the influence of the operating point, which
 239 Section 5 exploits: a change in the operating point alters the rates that
 240 turn the direction. Not every turn matters for a given indicator. Only the
 241 component of the motion along the great circle towards $\mathbf{v}$, the shortest path
 242 on the sphere between $\hat{\mathbf{d}}$ and $\mathbf{v}$, changes ψ ; the remaining $d - 2$ components
 243 move the direction without affecting the information retained by the indicator
 244 (Fig. 3(b)).

245 3.2. A bound in terms of the rotation arc

246 Suppose that over a window of the service life the informative direction
 247 traverses a path of total length $\Phi < \pi$ on the sphere, called its arc. The
 248 direction at the midpoint of this arc lies within $\Phi/2$ of every point on it
 249 (Fig. 3(c)), so a fixed indicator placed at the midpoint satisfies $\psi \leq \Phi/2$
 250 throughout. Since the loss increases with ψ , such an indicator satisfies, at
 251 every instant,

$$e^{2\ell} \leq \frac{1}{1 - w \sin^2(\Phi/2)} \leq \frac{1}{\cos^2(\Phi/2)}. \quad (12)$$

252 The same holds for any fixed indicator whose largest angle to the path, $\psi_{\max}$,
 253 is at most $\Phi/2$. The bound requires only the arc, which can be estimated
 254 from run-to-failure data before any indicator is designed. An arc of 30°
 255 allows at most 7.2% additional error variance, 60° at most 33%, and 90° a

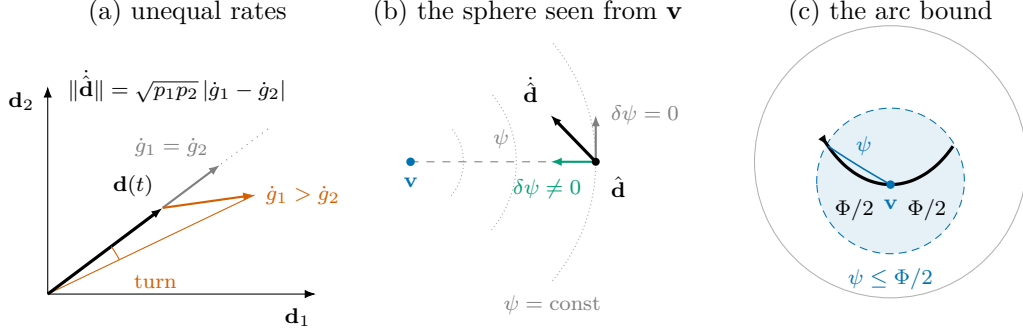


Figure 3: Mechanism of the rotation and of the arc bound. (a) The signal vector $\mathbf{d}$ of (5) for $d = 2$: when both components grow at the same logarithmic rate the direction $\hat{\mathbf{d}}$ is unchanged, and it turns only when the rates differ, at the speed (11). (b) A sector of the unit sphere seen from $\mathbf{v}$: the sets of constant ψ , and hence of constant loss, are circles around $\mathbf{v}$, and the great circles through $\mathbf{v}$ are radial lines. Only the component of $\hat{\mathbf{d}}$ along the great circle through $\mathbf{v}$ changes ψ ($\delta\psi \neq 0$); the other $d - 2$ components move $\hat{\mathbf{d}}$ within its level set ($\delta\psi = 0$). (c) A path of $\hat{\mathbf{d}}$ of length Φ , split at its midpoint into two halves of length $\Phi/2$. Every point of the path lies within $\Phi/2$ of the midpoint along the path, and therefore on the sphere, so an indicator $\mathbf{v}$ at the midpoint keeps $\psi \leq \Phi/2$ (shaded cap), which gives (12). Panels (b) and (c) hold in any dimension and are drawn for $d = 3$.

256 doubling; the right-hand bound corresponds to the limit of high signal-to-
 257 noise ratio and is independent of γ . Correspondingly, the safety margin of the
 258 replacement policy of Section 2.3 grows by at most the factor $1/\cos(\Phi/2)$.
 259 The best fixed indicator lies close to the midpoint of the arc; Supplementary
 260 Section S2 characterises it as a barycentre of the path of the informative
 261 direction. Figure 4(a) shows how the loss grows with the misalignment angle,
 262 and Fig. 4(b) the rotation measured on the battery cells of Section 4.

263 3.3. Design procedure

264 The bound yields a design procedure that requires run-to-failure data
 265 only.

- 266 1. Estimate the informative direction from the fleet: at each value of the
 267 normalised life τ , take the time derivative of the monitoring channels,
 268 whiten each channel by its measurement noise, estimated from the scatter
 269 of the channel about a smoothed trend, and average the resulting
 270 unit vectors over the units. The result is called the observed direction.
- 271 2. Compute the arc Φ that the observed direction traverses and the bound
 272 (12). The bound is the largest increase in error variance that a well-

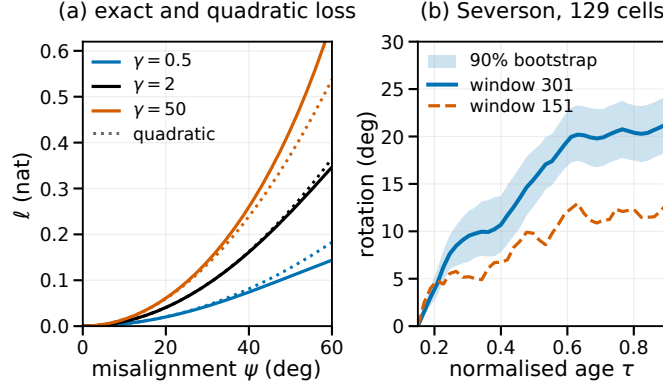


Figure 4: (a) The exact loss (7) (solid) and its quadratic approximation (dotted); the approximation overstates the loss when $\gamma < 2$ and understates it when $\gamma > 2$. (b) Rotation of the observed direction on the Severson battery cells (Section 4), estimated with smoothing windows of 301 and 151 cycles.

- placed fixed indicator can cause; if it is small compared with the accuracy the maintenance decision requires, a fixed indicator suffices.
3. Place the fixed indicator at the barycentre of the path, in practice the renormalised mean of the observed direction, and check that its largest angle to the path does not exceed $\Phi/2$.
4. Validate the indicator with a leave-one-out remaining-life estimator, against the same estimator run on all channels.
5. Evaluate the maintenance decision that the estimate supports.

Section 4 carries out the five steps on measured lithium-ion cells.

4. Case study: lithium-ion cells

The procedure is applied to the lithium-ion cells of Severson et al. [32], a public fleet cycled to failure under different fast-charging protocols and widely used as a prognostics benchmark. The monitoring channels are discharge capacity, internal resistance and mean cell temperature. The 129 cells that meet the selection criteria of the protocol in Supplementary Section S1 have lives from 447 to 1932 cycles; because the lives differ more than fourfold across protocols, the population is far from a time shift. The cells therefore violate (A1), and the procedure treats them as if they differed only in age along a common path in normalised life; the leave-one-out test of step 4 checks whether that approximation is good enough in practice. (A3) and

(A4) are not needed here. Supplementary Section S1 gives the protocol and parameters of every step.

4.1. Steps 1 to 3: rotation, bound and indicator

The observed direction of step 1, which estimates the informative direction as a fleet mean, moves through an arc of 30.7° between 15% and 90% of life (Fig. 4(b)). After whitening, the fixed indicator at its barycentre weights capacity, resistance and temperature as (0.99, 0.10, 0.12) and lies at most 14° from the informative direction, within half the arc. By (12), it therefore inflates the error variance of the remaining-life estimate by at most 7.5% over that window, irrespective of the signal-to-noise ratio, and widens the safety margin of a predictive replacement policy by at most 3.7%.

4.2. Step 4: remaining-life estimation

The bound concerns the optimal estimate given the record at one inspection. A practical estimator uses a history of inspections and is not optimal, so its error with the fixed indicator can fall on either side of its error with all channels; what the comparison tests is whether the information loss matters for such an estimator. A similarity-based estimator matches the last 30% of a cell's elapsed cycles against the fleet reference path in normalised life and returns the life that fits best; everything it uses, including the fixed indicator, is computed without the cell being tested. With all three channels its root-mean-square error is 148, 83 and 49 cycles at 50%, 70% and 90% of life. Run on the fixed indicator at the barycentre of the observed direction, the same estimator attains between 0.96 and 1.00 times the mean squared error of the three-channel estimator at these ages (Table 1). Capacity alone, the conventional battery health indicator, lies up to 22° from the informative direction, for which the bound allows 17%, and raises the mean squared error by 5% at half of life and by less later. Temperature alone, a deliberately misaligned control, raises it at least 40-fold, so the test is able to detect a poorly chosen indicator. A data-level fusion indicator, whose weights minimise the between-cell spread of the indicator at failure relative to its mean change over life, in the spirit of [8], lies 75° to 82° from the informative direction and has 28.9 times the mean squared error at half of life. Near failure, the stage its construction targets, it is the most accurate signal, at 0.81 times the three-channel error: it suppresses between-cell differences in the level of the channels at failure, a source of error that a time-shift population does not contain.

Table 1: Case study on the Severson cells. $\psi_{\max}$ is the largest angle between the estimator input and the fleet-mean informative direction over $\tau \in [0.15, 0.90]$; the mean-squared-error (MSE) ratios compare the leave-one-out remaining-life estimator run on the input with the same estimator run on all three channels, at 50%, 70% and 90% of life; the replacement columns give the minimal long-run cost per 1000 cycles, in units of a preventive replacement, when a failure costs ten times as much, with the fraction of cells failing and the mean life left unused at replacement, in cycles.

estimator input	$\psi_{\max}$	MSE ratio			replacement policy		
		50%	70%	90%	cost	failures	unused
all channels	—	1	1	1	1.689	1.6%	134
fixed indicator	14°	0.960	0.972	0.999	1.688	1.6%	133
capacity	22°	1.050	1.014	0.986	1.690	1.6%	134
fusion	82°	28.9	8.43	0.807	1.731	1.6%	150
temperature	90°	43.7	171	725	—	—	—

329 4.3. Step 5: replacement decisions

330 The last step evaluates the decision directly. Each cell is inspected every
331 25 cycles from cycle 100 and is replaced at the first inspection at which
332 the predicted remaining life is at most a threshold of n_{thr} cycles; a cell not
333 replaced before its end of life counts as a failure. For a failure ten times as
334 costly as a preventive replacement, the threshold that minimises the long-run
335 cost per cycle [29] gives the costs of Table 1. The fixed indicator, capacity
336 alone and all three channels reach the same cost to within 0.1%, with 1.6%
337 of cells failing and about 134 cycles, a fifth of a cell’s life, left unused at
338 replacement; the fusion indicator costs 2.5% more. For cost ratios from 5
339 to 50 the fixed indicator stays within 1% of the three-channel cost. The
340 thresholds are chosen on the same cells for every input, so the absolute costs
341 are optimistic, but the comparison between inputs is not. The case study
342 thus confirms, on measured data and at the level of a maintenance decision,
343 what the bound predicts: for this fleet a fixed indicator is adequate.

344 5. Selecting the operating point

345 The case study took the operating conditions as given. In many applica-
346 tions they are not: operators choose temperatures, loads or charging rates,
347 often to extend life. Because the operating point sets the relative rates of the
348 mechanisms, it also moves the informative direction, and it could in principle
349 be chosen to keep a fixed indicator aligned. This practice, referred to here as

steering, is examined in this section. No public fleet lets the operating point act on the damage (Section 6), so the examination uses a synthetic system. Section 3 already limits what steering can achieve: the excess error variance that it can remove is at most the bound (12) for the arc of the unsteered trajectory.

5.1. Limits of steering

At a state x , the directions attainable through the operating point form the set

$$\mathcal{D}(x) = \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\}, \quad (13)$$

and the smallest misalignment achievable at x , called the floor, is $\beta_{\min}(x) = \angle(\mathbf{v}, \mathcal{D}(x))$. Two structural results, derived in Supplementary Section S4, delimit what steering can do. First, holding the misalignment at a constant, non-zero level is a single scalar condition and needs only one adjustable operating variable. Second, removing the loss altogether is far more demanding. If the informative direction equals $\mathbf{v}$ at every instant, the damage can only grow along the fixed direction $\Sigma^{1/2}\mathbf{v}$, so the state must move along a straight line in damage space. Keeping it on that line requires, except for special combinations of parameters, one fewer independent operating variable than there are mechanisms, together with an aligning input that stays within the allowed envelope $\mathcal{U}$ along the whole line. A component with three mechanisms and a single temperature setpoint can therefore reduce the loss but, generically, not remove it.

When the loss cannot be removed, the relevant quantity is the smallest achievable average loss,

$$\bar{\ell}^* = \min_{u(\cdot)} \frac{1}{T_f} \int_0^{T_f} \ell \, dt, \quad (14)$$

over all operating policies $u(\cdot)$, that is, all ways of setting the operating point over time. Both the integral and the lifetime T_f depend on the policy, so the objective is a ratio of two quantities that the operating point changes together, and the operating point that aligns the indicator is in general not the one that prolongs life. The simplest policy disregards the lifetime and applies, at each state, the input that minimises the misalignment; it is referred to as the myopic policy.

5.2. A numerical study

The synthetic system is chosen to resemble the thermally activated degradation of a battery with three mechanisms and one adjustable temperature. It has $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is a scaled reciprocal temperature, $u = 3(1/(kT) - 1/(kT_{60})) / (1/(kT_{25}) - 1/(kT_{60}))$ with T_{60} and T_{25} the absolute temperatures of 60 and 25 °C, so that $u = 0$ corresponds to 60 °C and $u = 3$ to 25 °C. The parameters are $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, which in these units are activation energies of 0.44, 0.81 and 0.59 eV, with shape functions $h_i(x) = 1 + c_i x$ and $c = (-0.5, 2.0, 0.8)$, which give one decelerating and two accelerating mechanisms. Time is measured in units of the characteristic time of the first mechanism at 60 °C, where $\varphi_1 = 1$. There is a single input $u \in [0, 3]$, so by Section 5.1 the loss can be reduced but, generically, not removed; Supplementary Section S4 confirms this for the present system. The noise has unit variance and the prior variance is $\text{Var}[R] = 10^3$; the weight $w = \gamma/(1+\gamma)$ of (7) is evaluated at the system’s own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$; the values of the reported constants at $x_f = 1$ are given in Supplementary Section S4. The identities of Sections 2 and 3 were verified numerically on this system (Supplementary Section S3).

The bound of Section 3 can be checked first. Under constant inputs the arc is between 24.8° and 35.6°, the bound allows between 4.8% and 10.3%, and the best fixed indicator for each input incurs between 1.6% and 3.4%; the bound therefore holds and overestimates the excess by a factor of about three. These small excesses are all that steering can remove.

The comparison of strategies needs three indicators. The steering indicator q^* is the one for which the myopic policy achieves the smallest average loss; the balanced indicator weights the three whitened channels equally; and for each constant operating point, the constant design uses the indicator that is best for that point. With the input of the myopic policy searched on a grid of 41 points of $\mathcal{U}$, the steering indicator has the weight vector $\mathbf{v}^* = (0.466, 0.771, 0.435)$, at which the myopic policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. This comparison holds the indicator fixed at the one designed for steering. For operation without steering, the indicator would instead be designed for the constant input, and minimising over the indicator and the constant input jointly gives 7.95×10^{-3} , at the high-temperature end $u = 0$ of the envelope.

Relative to this baseline, the reduction achieved by steering is a factor of 47, the figure quoted in the abstract. The myopic policy is also close to the best achievable by any policy: the best policy for the ratio (14), computed on state grids of up to 137 points per axis, lies within 2.1% of it on every control grid, and both factors are stable under refinement of the grids (Supplementary Section S4).

Table 2 reports the practical consequences. By (8), a loss of 7.95×10^{-3} already corresponds to an error variance only 1.6% above that of the complete record, and steering removes almost all of this excess. A practical estimator, however, is largely insensitive to it. A time-shifted population is simulated along the path of each policy, each unit is observed at 30 instants over the last 40% of a lifetime before its remaining life is estimated, with unit channel noise, and the estimate is the time offset that best matches the reference path of the policy, computed once from the indicator and once from all three channels. Under every policy the two estimators differ by less than 1.2%, and the error as a fraction of life stays between 8.4% and 10.7%, being determined by the path rather than by the indicator. The operating point does, however, change the lifetime substantially. Across the constant designs of the table it varies 19-fold, because the mechanisms run faster when hot. Among the policies, the myopic policy at q^* attains a lifetime 1.66 times that of the best constant input at the same indicator and 3.8 times that of the constant design with the smallest loss; and a myopic policy aimed at the balanced indicator, which requires far less steering, extends the lifetime by a further factor of 3.4. For this system the choice of operating point matters far more for lifetime than for monitoring.

5.3. Sensitivity to model misspecification

Where run-to-failure data are scarce, the indicator can instead be designed from a degradation model, by applying steps 1 to 3 of the procedure to trajectories simulated from the model. Such an indicator inherits the errors of the model, and this subsection asks how accurate the model must be. For each error level ω , a perturbed energy vector $E(1 + \omega\eta)$ is drawn, with η standard normal, and the barycentre indicator is computed from trajectories of this misspecified system under a constant input. The indicator is then evaluated on the true system under the myopic policy aimed at it, with every loss evaluated at the system's own γ . Since the policy is computed from the true system, the table isolates the effect of an error in the indicator; an error that also affected the policy would be more costly. Table 3 reports the

Table 2: Information loss, remaining-life estimation error and lifetime of the synthetic system under different operating strategies. $\bar{\ell}$ is the information loss (9); $e^{2\bar{\ell}} - 1$ the corresponding excess error variance (8); the remaining-useful-life (RUL) errors are the median absolute error of a template-matching remaining-life estimator over 2000 units, as a percentage of life, based on the indicator or on all three channels; T_f is the lifetime. Constant designs use the indicator that is best for their own input.

operating strategy	$\bar{\ell}$	$e^{2\bar{\ell}} - 1$	RUL error (% of life)		
			indicator	all channels	T_f
constant, 60 °C	7.95×10^{-3}	1.6%	10.7%	10.8%	0.378
constant, 41.5 °C	1.44×10^{-2}	2.9%	8.9%	8.8%	1.967
constant, 25 °C	1.26×10^{-2}	2.6%	8.8%	8.7%	7.203
myopic, q^*	1.68×10^{-4}	0.03%	9.5%	9.5%	1.433
myopic, balanced	4.08×10^{-3}	0.8%	8.4%	8.5%	4.928

Table 3: Effect of designing the indicator from a misspecified stress model. Entries are the loss ratio: the loss on the true system relative to that of the indicator designed from the true model, over 40 draws per level; ω is the relative standard deviation of the error in the activation energies. The steering policy uses the true system.

ω	median	90th percentile	worst of 40
0.02	1.03	1.52	2.04
0.05	1.40	2.94	6.44
0.10	3.10	10.8	67.7
0.20	10.0	44.9	78.6
0.35	36.0	94.1	195

loss ratio, the loss of this indicator divided by that of the indicator designed from the true model, over 40 draws per level; a ratio of 1 indicates that the error has no effect.

An error of 2% in the energies has a negligible effect. At 5% the median loss ratio is 1.40. At 10% the median loss ratio is 3.10 and the worst of forty draws reaches 67.7, comparable in magnitude to the entire benefit of steering; beyond this level every column increases monotonically, to a median of 36.0 at $\omega = 0.35$. The loss to which these ratios apply is small, however, so that even a tenfold increase leaves the indicator’s excess error variance below 1%.

6. Evidence from public run-to-failure data

The preceding sections rest on premises that public run-to-failure data can partly test. Three are examined here: that the informative direction

467 rotates; that the operating point moves it by acting on the degradation, the
 468 actuation premise on which Section 5 relies; and that the separable stress
 469 model (A4), interpreted through the pairing of channels with mechanisms
 470 (A3), holds. The tests use the observed direction of step 1 of the procedure,
 471 estimated from data, which coincides with the informative direction only
 472 under (A1) and (A3). Table 4 summarises what six public run-to-failure
 473 fleets establish and what they leave open. The fleets are the Severson battery
 474 cells, the XJTU-SY and PRONOSTIA bearing test rigs, the NASA battery
 475 cells, and the C-MAPSS and N-CMAPSS turbofan engine simulations of
 476 NASA. Supplementary Section S1 gives the protocol behind every row, and
 477 its Table S1 lists the parameters of the code that produced it.

478 Rotation of the informative direction is observed on the battery cells
 479 of Section 4. That analysis and the turbofan contrasts below interpret a
 480 time derivative as the sensitivity (3), and therefore presuppose the time-shift
 481 population (A1).

482 The effect of the operating point on degradation remains untested. The
 483 measured fleets either do not vary the operating point within a unit or con-
 484 tain too few units. The two turbofan simulations do vary it, between discrete
 485 operating regimes of altitude, speed and throttle, but their generators pre-
 486 clude the test. C-MAPSS imposes its flow and efficiency losses as exponential
 487 functions of time, from a random initial deterioration, and lets the regime
 488 act only on the thermodynamic response of the sensors [36]. N-CMAPSS im-
 489 poses a linear normal degradation followed by a steeper abnormal one, with
 490 the operating history entering through the onset of the abnormal phase [38],
 491 and the health parameters supplied with the data are constant within every
 492 flight. On C-MAPSS the observed directions of different regimes of the same
 493 engine differ by more (B) than those of two halves of one regime (W). This
 494 contrast, positive in all but three engines against a negative control that re-
 495 turns none, therefore shows that the observed direction depends on the regime
 496 through the sensor response alone. Variable operating conditions thus rotate
 497 the observed direction even when they do not act on the damage. Because
 498 the loss of a fixed indicator depends on the observed direction regardless of
 499 the cause of its motion, this finding is relevant to indicator design; it is not,
 500 however, evidence that the operating point acts on the damage.

501 The separable model fails on raw turbofan sensors. On both N-CMAPSS
 502 datasets the interaction between operating point and life stage exceeds the
 503 main effect of the operating point. The test rejects (A3) and (A4) jointly, and
 504 since the simulated damage does not respond to the operating point within

Table 4: Premises of the model examined on six public run-to-failure fleets. Only the two turbofan simulations vary the operating point within a unit, and neither allows it to act on the damage; the actuation premise is therefore untested on every public fleet. B and W are the mean angles between observed directions of different operating regimes and of two halves of the same regime.

fleet	premise	quantity measured	finding
Severson lithium-ion [32]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 4(b)); all three weights stay positive once the smoothing window reaches 301 cycles
XJTU-SY [33], PRONOS-TIA [34]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [35]	actuation	—	only four cells
C-MAPSS [36]	actuation: not testable	between-regime minus within-regime angle, $B - W$, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; surrogate regimes give $ B - W \leq 0.5^\circ$; the damage is independent of the regime by construction, so the contrast reflects the sensor map
N-CMAPSS DS02, DS01 [37]	separability with pairing (4)	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected jointly

505 a flight, the violation is attributable to the sensor map: the sensors mix
 506 mechanisms with weights that depend on the operating point, in violation
 507 of (4). The pairing is what makes the informative direction positive, so the
 508 geometry of Sections 3 and 5 applies to such sensors only after the observation
 509 map has been inverted, and the offline screen of how far the operating point
 510 can move the informative direction (its actuation authority; Supplementary
 511 Section S4) is available only where the separable form has been shown to hold.
 512 The two-way layout, whose protocol is given in Supplementary Section S1,
 513 provides this test at the cost of a single pass over data that a reliability
 514 programme typically already holds.

515 7. Discussion and conclusions

516 The paper asked how much information about remaining life a fixed health
 517 indicator discards when degradation mechanisms progress at different rates,
 518 and whether the operating point can reduce the loss. The loss is set by
 519 how far the informative direction turns over the life, and on the battery
 520 cells examined here the turn is small enough that a fixed indicator supports
 521 remaining-life estimates and replacement decisions as well as all channels do.
 522 The operating point can reduce the loss, but when the turn is small there is
 523 little to gain.

524 The results have three practical implications.

525 First, the rotation arc should be assessed before an adaptive indicator
 526 is considered. The bound (12) converts a measured arc into a worst-case
 527 increase in the error variance of the remaining-life estimate. For the battery
 528 fleet examined here the resulting increase is at most 7.5%, and for the syn-
 529 thetic system the actual excess is about one third of the bound. Where the
 530 arc amounts to a few tens of degrees, a single fixed indicator is adequate;
 531 indicators whose weights change with age, whether adaptive, segmented by
 532 life stage or learned, become worthwhile as the arc approaches a right angle,
 533 at which the bound allows the error variance to double. The five steps of
 534 Section 3.3 make this assessment from run-to-failure data alone.

535 Second, the operating point should be selected primarily for service life.
 536 Steering the operating point can reduce the information loss by an order of
 537 magnitude or more. Where the loss is already small, however, this yields
 538 almost no improvement in the accuracy of a practical remaining-life estima-
 539 tor, whereas the same choice alters the lifetime by an order of magnitude.
 540 Aligning an indicator and prolonging the life of the component are distinct
 541 objectives, and when the arc is small the latter is the more consequential.

542 Third, variable operating conditions should be represented when the arc
 543 is estimated. On the turbofan simulations the observed direction differs
 544 between operating regimes even though the damage does not respond to
 545 them, because the sensors respond differently in different regimes. An arc
 546 estimated under a single operating condition can therefore understate the
 547 rotation that a fixed indicator will meet in service.

548 *Limitations.* Four limitations qualify these conclusions. First, the construc-
 549 tion assumes a time-shift population (A1); dispersing the degradation coeffi-
 550 cients of the synthetic system rotates the population regression of damage on

551 remaining life by about 45° away from the direction the construction uses,
552 more than the measured arc of the battery cells, so the bound does not cover
553 such dispersion, although the battery case study, whose cells violate (A1),
554 suggests that the practical conclusion survives it. Second, the pairing of
555 channels with mechanisms (A3) fails on raw turbofan sensors, so the geome-
556 try applies to them only after the observation map has been inverted. Third,
557 the actuation premise remains untested on every public fleet. Finally, the
558 best achievable loss is computed and bracketed between the myopic policy
559 and an extrapolation of the computation to infinitely fine grids, not bounded
560 analytically.

561 Appendix A. Derivations

562 Appendix A.1. Information loss and error-variance inflation

563 Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$, the linearised whitened record.
564 The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal
565 because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$. The record's sufficient
566 statistic $\mathbf{d}^\top \tilde{y}$ has ratio γ . A scalar Gaussian channel of signal-to-noise ratio
567 s carries $\frac{1}{2} \log(1 + s)$ nats about R [39], so the loss, the mutual information
568 of R with the record minus that with the indicator, is the first form of (7);
569 substituting $\cos^2 \psi = 1 - \sin^2 \psi$ gives the second. The posterior variances are
570 $\text{Var}[R]/(1 + \gamma \cos^2 \psi)$ and $\text{Var}[R]/(1 + \gamma)$, whose quotient is (8). Equation (10)
571 is (7) with $\gamma = \text{Var}[R]I_{\text{rec}}$ and $\gamma \cos^2 \psi = \text{Var}[R]I_z$; at fixed I_{rec} , w is fixed
572 and ℓ increases continuously from zero towards $\frac{1}{2} \log(1 + \gamma)$ as ψ goes from
573 0 to $\pi/2$.

574 Appendix A.2. Rotation of the informative direction

575 From $\mathbf{d}_i = e^{g_i}$, $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, and $\dot{\mathbf{d}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has components
576 $\hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$ with $p_i = \hat{\mathbf{d}}_i^2$, so its squared norm is $\text{Var}_p[\dot{g}]$. Differentiating
577 $g_i = \log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (1) gives $\dot{g}_i$ in (11). With $\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v}$,
578 $-\sin \psi \dot{\psi} = \dot{\mathbf{d}}^\top \mathbf{v}$, so only the component of $\dot{\mathbf{d}}$ along the great circle towards
579 $\mathbf{v}$ changes ψ .

580 Data availability and reproducibility

581 The six fleets named in Section 6 are publicly available. One script re-
582 computes every reported number from a single configuration and compares it

583 with the value stated in this manuscript and in its supplementary material,
584 covering the tables and the quantities reported alongside them; all of its
585 checks pass, and the output of the run that verified this version is included.
586 The random-system sweep and the downstream repetition of Supplemen-
587 tary Section S4 and the grid sequence of Section 5.2 require several hours
588 of computation, so they are recorded by three further scripts; the checks
589 read these records and recompute part of the grid sequence afresh. One
590 group of numbers lies outside the checks: the residuals of the identity checks
591 in Supplementary Section S3. The figures are produced by three further
592 scripts. All of these are available at [https://github.com/HuyHoangLe0201/](https://github.com/HuyHoangLe0201/health-indicator-information-loss)
593 [health-indicator-information-loss](https://github.com/HuyHoangLe0201/health-indicator-information-loss), and are also provided with this ar-
594 ticle as a supplementary archive, together with the Supplementary Material.

595 **CRedit authorship contribution statement**

596 **Huy Hoang Le:** Conceptualization, Methodology, Formal analysis, Soft-
597 ware, Validation, Investigation, Data curation, Visualization, Writing – orig-
598 inal draft. **Kim-Anh Nguyen:** Conceptualization, Methodology, Valida-
599 tion, Writing – review & editing, Supervision, Project administration.

600 **Declaration of generative AI and AI-assisted technologies in the** 601 **manuscript preparation process**

602 During the preparation of this work the authors used Claude (Anthropic)
603 to assist with drafting and editing the text and with writing and checking
604 the analysis and verification code. After using this tool, the authors reviewed
605 and edited the content as needed and take full responsibility for the content
606 of the published article.

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608 The authors declare that they have no known competing financial inter-
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References

- [1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, J. Lin, Machinery health prognostics: A systematic review from data acquisition to RUL prediction, *Mech. Syst. Signal Process.* 104 (2018) 799–834. doi:10.1016/j.ymssp.2017.11.016.
- [2] J. Coble, J. W. Hines, Identifying optimal prognostic parameters from data: A genetic algorithms approach, *Annual Conference of the PHM Society*, vol. 1, no. 1, San Diego, CA (2009).
- [3] H. H. Le, K.-A. Nguyen, Information-theoretic limits of health indicator construction for remaining useful life prediction, *Eng. Sci. Technol. Int. J.* 81 (2026) 102485. doi:10.1016/j.jestch.2026.102485.
- [4] F. Karimi Pour, D. Theilliol, V. Puig, G. Cembrano, Health-aware control design based on remaining useful life estimation for autonomous racing vehicle, *ISA Trans.* 113 (2021) 196–209. doi:10.1016/j.isatra.2020.03.032.
- [5] M. S. Félix, J. J. Martinez, C. Bérenguer, Remaining useful life (RUL) control of controlled systems under degradation, *Int. J. Robust Nonlinear Control* 35 (18) (2025) 8095–8116. doi:10.1002/rnc.70105.
- [6] J. C. Salazar, P. Weber, F. Nejari, R. Sarrate, D. Theilliol, System reliability aware model predictive control framework, *Reliability Engineering & System Safety* 167 (2017) 663–672. doi:10.1016/j.ress.2017.04.012.
- [7] P. Wen, S. Zhao, S. Chen, Y. Li, A generalized remaining useful life prediction method for complex systems based on composite health indicator, *Reliability Engineering & System Safety* 205 (2021) 107241. doi:10.1016/j.ress.2020.107241.
- [8] K. Liu, N. Z. Gebraeel, J. Shi, A data-level fusion model for developing composite health indices for degradation modeling and prognostic analysis, *IEEE Trans. Autom. Sci. Eng.* 10 (3) (2013) 652–664. doi:10.1109/TASE.2013.2250282.
- [9] C. Song, K. Liu, Statistical degradation modeling and prognostics of multiple sensor signals via data fusion: A composite health index approach, *IISE Trans.* 50 (10) (2018) 853–867. doi:10.1080/24725854.2018.1440673.

- 647 [10] X. Fang, K. Paynabar, N. Gebraeel, Multistream sensor fusion-
648 based prognostics model for systems with single failure modes,
649 Reliability Engineering & System Safety 159 (2017) 322–331.
650 doi:10.1016/j.ress.2016.11.008.
- 651 [11] Y. Wang, I.-C. Lee, Y. Hong, X. Deng, Building degradation index with
652 variable selection for multivariate sensory data, Reliability Engineering
653 & System Safety 227 (2022) 108704. doi:10.1016/j.ress.2022.108704.
- 654 [12] H. Li, Z. Zhang, X. Si, A dual-purpose data-model interactive frame-
655 work for multi-sensor selection and prognosis, Reliability Engineering &
656 System Safety 258 (2025) 110904. doi:10.1016/j.ress.2025.110904.
- 657 [13] H. Yan, K. Liu, X. Zhang, J. Shi, Multiple sensor data fusion for
658 degradation modeling and prognostics under multiple operational con-
659 ditions, IEEE Transactions on Reliability 65 (3) (2016) 1416–1426.
660 doi:10.1109/tr.2016.2575449.
- 661 [14] L. Guo, N. Li, F. Jia, Y. Lei, J. Lin, A recurrent neural network based
662 health indicator for remaining useful life prediction of bearings, Neuro-
663 computing 240 (2017) 98–109. doi:10.1016/j.neucom.2017.02.045.
- 664 [15] K. Javed, R. Gouriveau, N. Zerhouni, State of the art and tax-
665 onomy of prognostics approaches, trends of prognostics applica-
666 tions and open issues towards maturity at different technology
667 readiness levels, Mech. Syst. Signal Process. 94 (2017) 214–236.
668 doi:10.1016/j.ymssp.2017.01.050.
- 669 [16] N. Li, N. Gebraeel, Y. Lei, X. Fang, X. Cai, T. Yan, Re-
670 maining useful life prediction based on a multi-sensor data fusion
671 model, Reliability Engineering & System Safety 208 (2021) 107249.
672 doi:10.1016/j.ress.2020.107249.
- 673 [17] W. Q. Meeker, L. A. Escobar, Statistical Methods for Reliability Data,
674 Wiley, 1998.
- 675 [18] Z.-S. Ye, M. Xie, Stochastic modelling and analysis of degradation for
676 highly reliable products, Appl. Stoch. Models Bus. Ind. 31 (1) (2015)
677 16–32. doi:10.1002/asmb.2063.

- [19] X.-S. Si, W. Wang, C.-H. Hu, D.-H. Zhou, Remaining useful life estimation — a review on the statistical data driven approaches, *Eur. J. Oper. Res.* 213 (1) (2011) 1–14. doi:10.1016/j.ejor.2010.11.018.
- [20] Z. Zhang, X. Si, C. Hu, Y. Lei, Degradation data analysis and remaining useful life estimation: a review on Wiener-process-based methods, *Eur. J. Oper. Res.* 271 (3) (2018) 775–796. doi:10.1016/j.ejor.2018.02.033.
- [21] L. Bian, N. Gebraeel, J. P. Kharoufeh, Degradation modeling for real-time estimation of residual lifetimes in dynamic environments, *IIE Transactions* 47 (5) (2015) 471–486. doi:10.1080/0740817x.2014.955153.
- [22] N. Li, N. Gebraeel, Y. Lei, L. Bian, X. Si, Remaining useful life prediction of machinery under time-varying operating conditions based on a two-factor state-space model, *Reliability Engineering & System Safety* 186 (2019) 88–100. doi:10.1016/j.ress.2019.02.017.
- [23] S. Jahani, S. Zhou, D. Veeramani, Stochastic prognostics under multiple time-varying environmental factors, *Reliability Engineering & System Safety* 215 (2021) 107877. doi:10.1016/j.ress.2021.107877.
- [24] Z. Lin, X. Liu, Y. Xiang, Y. Hong, Modeling multivariate degradation data with dynamic covariates under a bayesian framework, *Reliability Engineering & System Safety* 261 (2025) 111115. doi:10.1016/j.ress.2025.111115.
- [25] S.-S. Chen, X.-Y. Li, Modeling dependent degradation data considering inherent causal relationships for reliability analysis and remaining useful life prediction, *Reliability Engineering & System Safety* 277 (2026) 113080. doi:10.1016/j.ress.2026.113080.
- [26] Y. Langeron, A. Grall, A. Barros, A modeling framework for deteriorating control system and predictive maintenance of actuators, *Reliability Engineering & System Safety* 140 (2015) 22–36. doi:10.1016/j.ress.2015.03.028.
- [27] M. A. uit het Broek, R. H. Teunter, B. de Jonge, J. Veldman, Joint condition-based maintenance and condition-based production optimization, *Reliability Engineering & System Safety* 214 (2021) 107743. doi:10.1016/j.ress.2021.107743.

- [28] J. Zuo, C. Cadet, Z. Li, C. Bérenguer, R. Outbib, A deterioration-aware energy management strategy for the lifetime improvement of a multi-stack fuel cell system subject to a random dynamic load, *Reliability Engineering & System Safety* 241 (2024) 109660. doi:10.1016/j.ress.2023.109660.
- [29] A. Kamariotis, K. Tatsis, E. Chatzi, K. Goebel, D. Straub, A metric for assessing and optimizing data-driven prognostic algorithms for predictive maintenance, *Reliability Engineering & System Safety* 242 (2024) 109723. doi:10.1016/j.ress.2023.109723.
- [30] H. H. Le, K.-A. Nguyen, Generalization bounds and sample complexity for remaining useful life prediction from complete degradation trajectories, *Meas. Sci. Technol.* 37 (2026) 226203. doi:10.1088/1361-6501/ae7109.
- [31] H. Lim, B.-J. Yum, Optimal design of accelerated degradation tests based on Wiener process models, *J. Appl. Stat.* 38 (2) (2011) 309–325. doi:10.1080/02664760903406488.
- [32] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H. Chen, M. Aykol, P. K. Herring, D. Fraggedakis, M. Z. Bazant, S. J. Harris, W. C. Chueh, R. D. Braatz, Data-driven prediction of battery cycle life before capacity degradation, *Nat. Energy* 4 (2019) 383–391. doi:10.1038/s41560-019-0356-8.
- [33] B. Wang, Y. Lei, N. Li, N. Li, A hybrid prognostics approach for estimating remaining useful life of rolling element bearings, *IEEE Trans. Rel.* 69 (1) (2020) 401–412. doi:10.1109/tr.2018.2882682.
- [34] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, C. Varnier, PRONOSTIA: An experimental platform for bearings accelerated degradation tests, in: *IEEE Int. Conf. Prognostics Health Manage. (PHM)*, Denver, CO, USA, 2012, pp. 1–8.
- [35] B. Saha, K. Goebel, Battery data set, NASA Ames Prognostics Data Repository (2007).
- [36] A. Saxena, K. Goebel, D. Simon, N. Eklund, Damage propagation modeling for aircraft engine run-to-failure simulation, in: *Int. Conf.*

- 742 Prognostics Health Manage. (PHM), Denver, CO, USA, 2008, pp. 1–
743 9. doi:10.1109/phm.2008.4711414.
- 744 [37] M. Arias Chao, C. Kulkarni, K. Goebel, O. Fink, Aircraft engine run-
745 to-failure dataset under real flight conditions for prognostics and diag-
746 nostics, Data 6 (1) (2021) 5. doi:10.3390/data6010005.
- 747 [38] M. Arias Chao, C. Kulkarni, K. Goebel, O. Fink, Fusing physics-based
748 and deep learning models for prognostics, Reliability Engineering & Sys-
749 tem Safety 217 (2022) 107961. doi:10.1016/j.ress.2021.107961.
- 750 [39] T. M. Cover, J. A. Thomas, Elements of Information Theory, 2nd Edi-
751 tion, Wiley, 2006. doi:10.1002/047174882x.